

SQL Fundamentals

Updated Exercises

Exercise 2 : SQL Aggregates and Operators

1. The students Table

01. List all distinct departments in the students table

```
SELECT DISTINCT department  
FROM students ;
```



department
IT
HR
Finance

02. Get the average age of students per department

```
SELECT department,  
AVG (age) as avg_age  
FROM students  
GROUP BY department ;
```



department	avg_age
IT	20.5
HR	22.0
Finance	23.0

03. Show departments with ^{more than} one student

```
SELECT department  
      COUNT(student_id) AS student_count  
FROM students  
GROUP BY department  
HAVING COUNT(student_id) > 1 ;
```



department	student_count
IT	2
HR	2

04. Get all students whose age is between 21 and 23

```
SELECT student_id,  
       name,  
       age,  
       department
```

```
FROM students
```

```
WHERE age BETWEEN 21 AND 23 ;
```



student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

05. List all students in the IT or HR department who are older than 21

```
SELECT student_id,  
       name,  
       age,  
       department
```

```
FROM students  
WHERE department IN ('IT', 'HR')  
AND age > 21;
```



student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

2. The courses Table

06. Show total credits per department, only for departments with more than 5 total credits

```
SELECT department,  
       SUM(credits) AS total_credits
```

```
FROM courses
```

```
GROUP BY department
```

```
HAVING SUM(credits) > 5;
```



department	total_credits
IT	11

07. List all courses that do not have 4 credits

```
SELECT course_id,  
        course_name,  
        department,  
        credits
```

```
FROM courses
```

```
WHERE credits <> 4;
```



course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

08. Show the top 3 courses by credits in descending order

```
SELECT course_id,  
        course_name,  
        credits
```

```
FROM courses
```

```
ORDER BY credits DESC
```

```
LIMIT 3;
```



Course id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

3. The enrollments Table

09. Get the maximum, minimum and average grade across all enrollments

```
SELECT MAX (grade) AS max_grade,  
       MIN (grade) AS min_grade,  
       AVG (grade) AS avg_grade  
FROM enrollments;
```



max_grade	min_grade	avg_grade
90	78	84.6

10. Count how many enrollments exist per course

```
SELECT course_id  
       COUNT (*) AS enrollment_count  
FROM enrollments  
GROUP BY course_id;
```



course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11. The salary Table

11. Find total salary and total bonus per department

```
SELECT department,  
       SUM (salary) AS total_salary,  
       SUM (bonus) AS total_bonus  
FROM salaries  
GROUP BY department;
```



department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. Show departments where average salary is above 55,000

```
SELECT department,  
       AVG (salary) AS avg_salary  
FROM salaries  
GROUP BY department  
HAVING AVG(salary) > 55000;
```



department	avg_salary
IT	61000
Finance	70000

B. List employees whose salary plus bonus is greater than 60,000

```
SELECT employee_id,  
       name,  
       salary,  
       bonus,  
       salary + bonus AS total_compensation  
FROM salaries  
WHERE salary + bonus > 60000;
```



employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

5. The projects Table

14. Show total and average budget per department. Only include departments with average budget above 70,000

```
SELECT department,  
       SUM(budget) AS total_budget,  
       AVG(budget) AS avg-budget  
FROM projects  
GROUP BY department  
HAVING AVG(budget) > 70000;
```



department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

15. List all projects with budget between 50000 and 120000, excluding the Marketing department

```
SELECT project_id,  
        project_name,  
        department,  
        budget
```

```
FROM projects
```

```
WHERE budget BETWEEN 50000 AND 120000
```

```
AND department <> 'Marketing';
```



project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000